NVLAP TESTING NVLAP LAB CODE:600177-0 ilac-MRA

GTG Global Testing Group

Test Report Of ANSI/IES LM-79-19

APPROVED METHOD FOR OPTICAL AND ELECTRICAL MEASUREMENTS OF SOLID-STATE LIGHTING PRODUCTS

Report Number..... : N02A23121042L02101

Client..... : RZ LIGHTING CO LTD

Address..... : 4F, Bldg. B, Zhongjing Sci-Tech Park, 606 Fengtang Blvd, Bao'an, Shenzhen, China 518103

Test Model..... : CP05-180L-[D;M;I;S;P;SI;SM;SP;SMP;SIP]-30, CP05-180L-[D;M;I;S;P;SI;SM;SP;SMP;SIP]-65

Brand Name..... : RZ

Testing Laboratory... : Guangdong Meide Testing Technology Co., Ltd.

Address..... : 1st floor, B Area, Jinbaisheng Industrial Park, Headquarters 2 Road,Songshan Lake Hi-tech Industrial Development Zone,Dongguan City,Guangdong Pr., China.

Testing Location..... : As above

Date of receipt..... : Dec. 25, 2023

Date of test ..... : Jan. 23, 2024 - Jan. 25, 2024

Date of report..... : Jan. 25, 2024

Tested by:	Checked by:	Approved by:
Allen chen	Jarvis zhang	Jessie Li
Allen Chen/ Test Engineer	Jarvis Zhang/ Project Engineer	Jessie Li/ Technical Manager

Note 1: The test data was only valid for the test sample(s).This test report is prepared for the customer shown above and for the device described herein. It may not be duplicated or use in part without prior written consent from Guangdong Meide Testing Technology Co., Ltd. This report must not be used by the customer to claim product certification, approval, or endorsement by NVLAP,NIST, or any agency of the Federal Government.
Note 2: This report does not imply product certification, approval, or endorsement by NVLAP, NIST,or any agency of the Federal Government.

1. Product Description for Equipment under Test(EUT)

Representative (Tested) Model:	CP05-180L-[D;M;I;S;P;SI;SM;SP;SMP;SIP]-30, CP05-180L-[D;M;I;S;P;SI;SM;SP;SMP;SIP]-65
All Model:	CP05-180L-[D;M;I;S;P;SI;SM;SP;SMP;SIP]-YY(YY can be 30, 40, 50, 57, 65 to denote CCT.)
Manufacturer:	RZ LIGHTING CO LTD
Product Type:	Fuel Pump Canopy Luminaires
Rated Voltage/Frequency:	120-277V AC, 50/60Hz
Rated Power:	180W/150W/120W
Rated luminous flux:	23400lm/19500lm/15600lm
Nominal CCT:	3000K/4000K/5000K/5700K/6500K
LED Manufacturer:	Lumileds Holding B.V.
LED Model No.:	L128-3080RA35002U1, L128-6580RA35002U1

2.Standards Used

- ANSI/IES LM-79-19:APPROVED METHOD:OPTICAL AND ELECTRICAL MEASUREMENTS OF SOLID-STATE LIGHTING PRODUCTS
- IES TM-30-18 IES Method for Evaluating Light Source Color Rendition (This Method is not in Nvlap accreditation scope)
- ANSI C82.77-10:2014 Harmonic Emission Limits – Related Power Quality Requirements for Lighting Equipment-Solid State

3.Test equipment list

Test Equipment	Serial No.	Model No.	Calibration due date
Full-field Speed Goniophotometer	MD-E028	GO-R5000	2024/09/16
Digital Power Meter	MD-E001	PF2010	2024/09/16
AC Testing Power Source	MD-E002	DPS1060	2024/09/16
Total Spectral Radiant Flux Standard Lamp	MD-E007	D908S	2024/09/25
Integrating Sphere System	MD-E029	2M	2024/09/16
High Accuracy Array Spectroradio Meter	MD-E011	HAAS-3000	2024/09/16
Digital Power Meter	MD-E008	PF310	2024/09/16
AC Testing Power Source	MD-E010	DPS1010	2024/09/16
Standard Lamp	MD-E036	D204	2024/09/25

Statement of Traceability: Guangdong Meide Testing Technology Co., Ltd. attested that all calibration has been performed using suitable standards traceable to national primary standards and International System of Unit(SI).

4. Test Method

Requirements of Ambient Condition
Product was tested with no seasoning. All stabilization and measurements were made in compliance with ANSI/IES LM-79-19. The product was operated at rated voltage or at voltage required by manufacturer. The ambient temperature of the sample was maintained at 25°C±1.2°C during measurement. And relative humidity between 10% and 65%.

Goniophotometer System
The sample was tested according to the ANSI/IES LM-79-19.

Photometric paramters were measured using a type C goniophotometer and software. The samples were operated at rated voltage and was stabilized before measurement. Luminous flux, Luminous efficacy, zonal flux were calculated from the software taken at 1° vertical intervals and 22.5°horizontal intervals.Photometric distance was more than five times of the Largest dimension of the test SSL product.

Integrating Sphere System
The sample was tested according to the ANSI/IES LM-79-19.

The sample measurements were made using a spectroradiometer connected by a fiber optic cable and detector through the detector port of the integrating sphere. Coating reflectance of the integrating sphere was 90% to 98%. Photometric measurement conditions was using 4π geometry. The self-absorption factor is applied in the final test result. The sample was operated at rated voltage and was stabilized before measurement. Chromaticity coordinates, correlated color temperature and color rendering index were calculated from the spectral radiant flux measurements taken at 1 nm intervals over the range of 380 to 780 nm.

Fidelity Index (R_f) and Gamut Index (R_g) Calculation
The R_f, R_g was calculated according to IES TM-30-18 by using calculation tools. The calculation was based on the measured SPD from 380nm to 780nm with 1nm intervals. All the colors in this report is for reference only.

THD and PF Test

The sample was tested according to the ANSI C82.77-10:2014.

The sample was operated at rated voltage and was stabilized before measurement. The total harmonic distortion were calculated from the digital power meter.

5. Integrating Sphere Test Results

5.1 Test Data

Test Ambient Temperature (Integrating sphere internal temperature)	25.3℃	Test orientation	Downward
Operate time(Min.)	60	stabilization time(Min.)	30

Optical and Electrical Measurement Result

Model	Voltage (V)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Luminous Flux(lm)	Efficacy (lm/W)	CCT (K)
CP05-180L-[D;M;I;S;P;SI;SM;SP;SMP;SIP]-30	119.98	60	1.512	181	0.9979	23753	131.21	2884
CP05-180L-[D;M;I;S;P;SI;SM;SP;SMP;SIP]-65	120.02	60	1.512	181.1	0.998	26749	147.7	6639

Model	Ra	R9	Rf	Rg	x	y	u'	v'	Duv
CP05-180L-[D;M;I;S;P;SI;SM;SP;SMP;SIP]-30	84.9	16	86	97	0.4417	0.3996	0.2556	0.5203	-2.40E-03
CP05-180L-[D;M;I;S;P;SI;SM;SP;SMP;SIP]-65	85.2	14	84	91	0.3096	0.3331	0.1941	0.4701	6.87E-03

5.2 Model # CP05-180L-[D;M;I;S;P;SI;SM;SP;SMP;SIP]-30 Color Rendering Index

Ra
84.9

R1	R2	R3	R4	R5
85	95	94	83	86

R6	R7	R8	R9	R10
94	82	61	16	88

R11	R12	R13	R14	R15
84	80	87	98	77

***5.3.1 Model # CP05-180L-[D;M;I;S;P;SI;SM;SP;SMP;SIP]-30 ANSI/IES TM-30-18 Color Rendition Report**

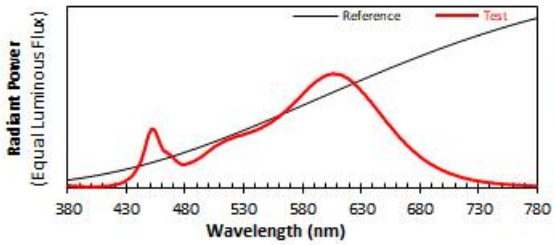
ANSI/IES TM-30-18 Color Rendition Report

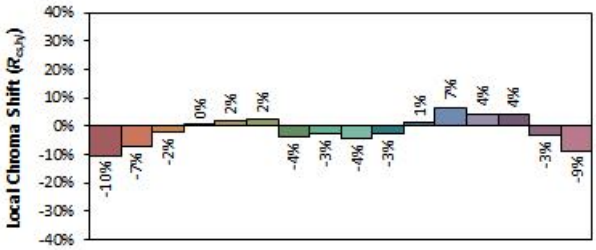
Source: L128-3080RA35002U1

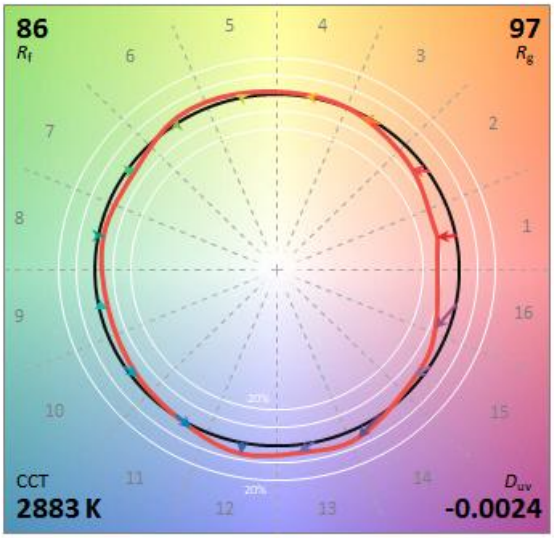
Manufacturer: RZ LIGHTING CO LTD

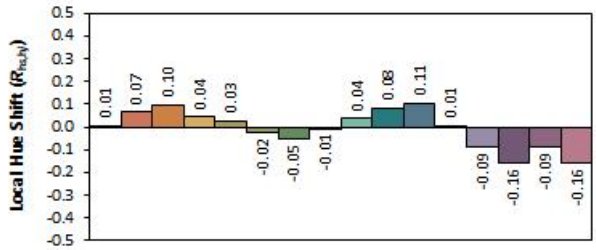
Date: 2024/1/22

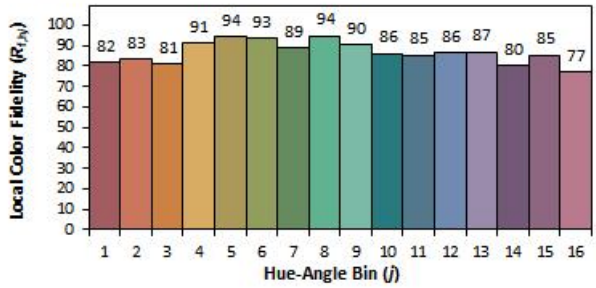
Model: CP05-180L-[D;M;I;S;P;SI;SM;SP;SMP;SIP]-30

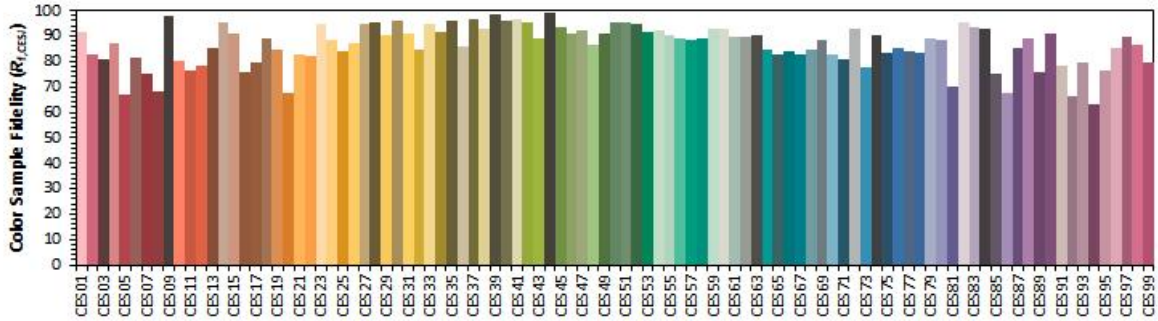












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.4417
y	0.3995
u'	0.2557
v'	0.5203

CIE 13.3-1995 (CRI)	
R_a	85
R_9	16

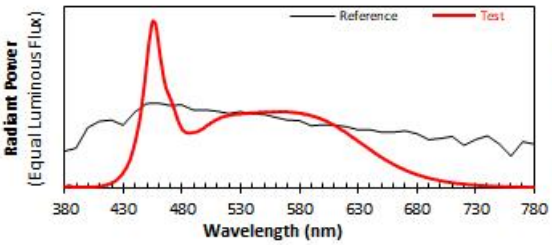
Colors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.00.

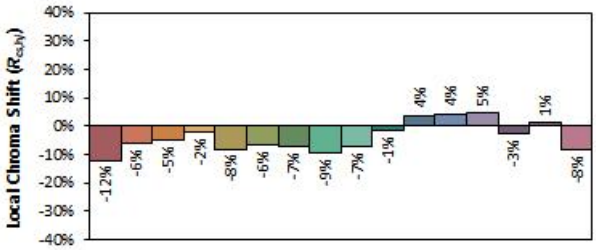
***5.3.2 Model # CP05-180L-[D;M;I;S;P;SI;SM;SP;SMP;SIP]-65 ANSI/IES TM-30-18 Color Rendition Report**

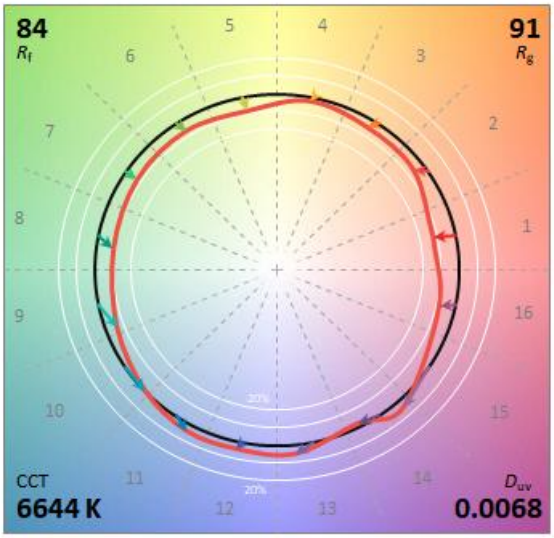
ANSI/IES TM-30-18 Color Rendition Report

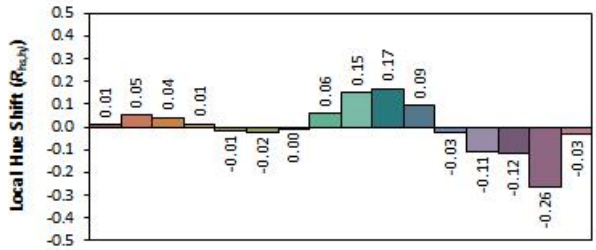
Source: L128-6580RA35002U1 **Manufacturer:** RZ LIGHTING CO LTD

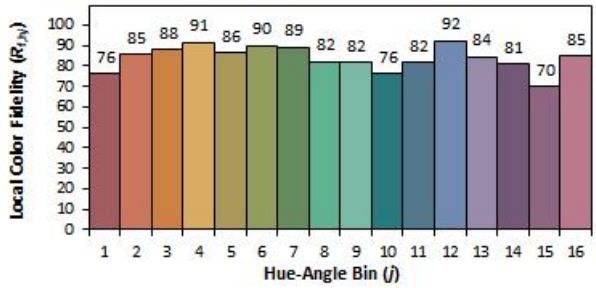
Date: 2024/1/25 **Model:** CP05-180L-[D;M;I;S;P;SI;SM;SP;SMP;SIP]-65

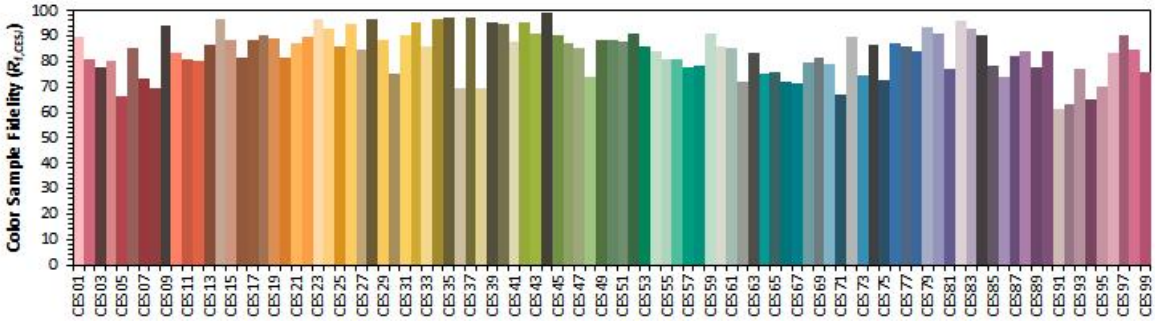












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

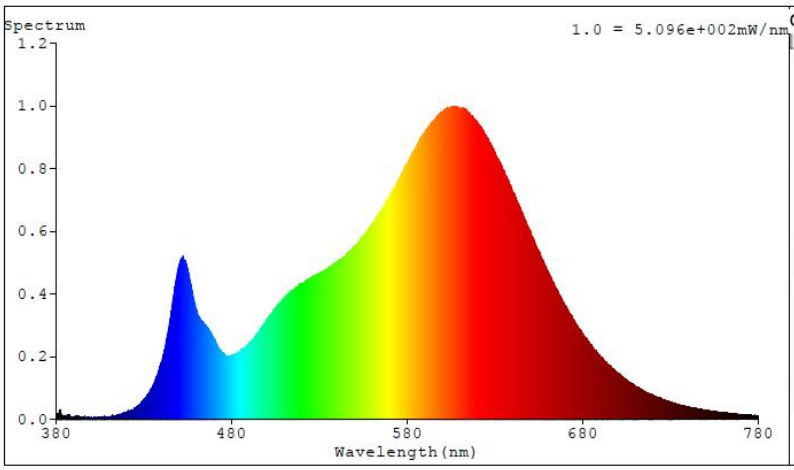
x	0.3095
y	0.3330
u'	0.1942
v'	0.4700

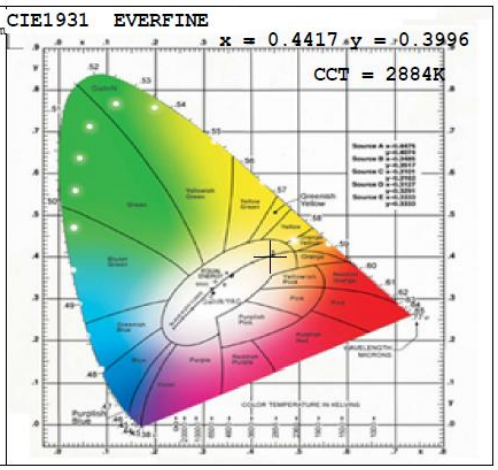
CIE 13.3-1995 (CRI)

R_a	85
R_9	14

Colors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.00.

5.4 Model # CP05-180L-[D;M;I;S;P;SI;SM;SP;SMP;SIP]-30 Relative Spectral Power Distribution





nm	mW	nm	mW	nm	mW	nm	mW	nm	mW
380	0.0118	414	0.0107	448	0.4341	482	0.205	516	0.4183
381	0.0203	415	0.01	449	0.4698	483	0.2129	517	0.4211
382	0.0126	416	0.0156	450	0.4852	484	0.2123	518	0.4243
383	0.012	417	0.0175	451	0.5089	485	0.22	519	0.4265
384	0.0067	418	0.019	452	0.5131	486	0.2251	520	0.4331
385	0.0137	419	0.0221	453	0.508	487	0.226	521	0.4357
386	0.009	420	0.0221	454	0.5018	488	0.2326	522	0.4412
387	0.0131	421	0.0235	455	0.482	489	0.238	523	0.4473
388	0.0086	422	0.028	456	0.4542	490	0.2459	524	0.4442
389	0.001	423	0.0296	457	0.4278	491	0.2508	525	0.4512
390	0.0079	424	0.0315	458	0.3965	492	0.2548	526	0.4512
391	0.0094	425	0.0385	459	0.3727	493	0.2642	527	0.4576
392	0.0113	426	0.04	460	0.3507	494	0.2704	528	0.4557
393	0.0102	427	0.0421	461	0.3286	495	0.2769	529	0.4631
394	0.0041	428	0.0485	462	0.3223	496	0.2842	530	0.4642
395	0	429	0.057	463	0.3104	497	0.2913	531	0.4706
396	0.004	430	0.0644	464	0.3102	498	0.3022	532	0.469
397	0.0046	431	0.0705	465	0.3007	499	0.3105	533	0.4787
398	0.0072	432	0.0772	466	0.2922	500	0.3166	534	0.4791
399	0.007	433	0.0847	467	0.2863	501	0.3261	535	0.4815
400	0.0068	434	0.0956	468	0.2786	502	0.3313	536	0.4849
401	0.0051	435	0.1041	469	0.2685	503	0.3396	537	0.4889
402	0.0011	436	0.1173	470	0.2588	504	0.3528	538	0.4965
403	0.0054	437	0.1329	471	0.2484	505	0.3553	539	0.4968
404	0.0022	438	0.1492	472	0.2335	506	0.3634	540	0.4996
405	0.0084	439	0.1679	473	0.2264	507	0.3685	541	0.5073
406	0.0077	440	0.1816	474	0.2178	508	0.3745	542	0.508
407	0.0069	441	0.2059	475	0.2126	509	0.3814	543	0.5145
408	0.0061	442	0.224	476	0.2072	510	0.3861	544	0.5185
409	0.0068	443	0.2529	477	0.2	511	0.3897	545	0.5229
410	0.0083	444	0.2885	478	0.2012	512	0.3987	546	0.5288
411	0.0093	445	0.3216	479	0.1994	513	0.4008	547	0.5364
412	0.012	446	0.3563	480	0.2035	514	0.4113	548	0.541
413	0.0091	447	0.3993	481	0.205	515	0.4117	549	0.5469

nm	mW	nm	mW	nm	mW	nm	mW	nm	mW
550	0.5495	599	0.9769	648	0.6266	697	0.1628	746	0.0336
551	0.5593	600	0.9798	649	0.6122	698	0.156	747	0.0331
552	0.5634	601	0.9856	650	0.6012	699	0.153	748	0.0329
553	0.5688	602	0.9887	651	0.5837	700	0.1481	749	0.031
554	0.5783	603	0.9924	652	0.5719	701	0.1442	750	0.0293
555	0.5831	604	0.9912	653	0.5599	702	0.1393	751	0.0293
556	0.59	605	0.9952	654	0.5478	703	0.1347	752	0.0278
557	0.6004	606	0.9983	655	0.5389	704	0.13	753	0.0268
558	0.61	607	1	656	0.5239	705	0.1265	754	0.0263
559	0.608	608	0.9925	657	0.5117	706	0.1224	755	0.0258
560	0.623	609	0.9929	658	0.501	707	0.1187	756	0.0242
561	0.6265	610	0.9935	659	0.4901	708	0.1137	757	0.0243
562	0.6414	611	0.9935	660	0.4724	709	0.1102	758	0.0232
563	0.6466	612	0.9897	661	0.465	710	0.1079	759	0.0226
564	0.6562	613	0.9887	662	0.4518	711	0.1032	760	0.0217
565	0.6657	614	0.984	663	0.4369	712	0.1021	761	0.0212
566	0.6729	615	0.9813	664	0.4279	713	0.0969	762	0.0215
567	0.6826	616	0.9697	665	0.4167	714	0.0941	763	0.0195
568	0.6918	617	0.9668	666	0.4086	715	0.0909	764	0.0193
569	0.7051	618	0.9634	667	0.3977	716	0.0879	765	0.0191
570	0.71	619	0.9545	668	0.384	717	0.0846	766	0.0185
571	0.7232	620	0.9487	669	0.3732	718	0.0829	767	0.0182
572	0.7314	621	0.9394	670	0.3651	719	0.0788	768	0.0176
573	0.7419	622	0.931	671	0.3545	720	0.078	769	0.0171
574	0.752	623	0.9265	672	0.3455	721	0.075	770	0.0164
575	0.7643	624	0.9145	673	0.3354	722	0.0734	771	0.0153
576	0.77	625	0.8996	674	0.3253	723	0.0712	772	0.0154
577	0.7842	626	0.8947	675	0.3167	724	0.0686	773	0.0148
578	0.7924	627	0.8836	676	0.3079	725	0.0656	774	0.0149
579	0.8041	628	0.8733	677	0.3005	726	0.0628	775	0.0151
580	0.8148	629	0.8652	678	0.2903	727	0.0617	776	0.0138
581	0.8277	630	0.8519	679	0.2828	728	0.0593	777	0.0134
582	0.8382	631	0.8377	680	0.2721	729	0.0583	778	0.0134
583	0.8475	632	0.8259	681	0.266	730	0.0564	779	0.0131
584	0.8566	633	0.8212	682	0.2565	731	0.0552	780	0.0132
585	0.8684	634	0.8087	683	0.2486	732	0.0516		
586	0.8815	635	0.7902	684	0.2406	733	0.0504		
587	0.8881	636	0.7788	685	0.2368	734	0.0494		
588	0.896	637	0.7674	686	0.2278	735	0.0481		
589	0.9064	638	0.7552	687	0.2214	736	0.0462		
590	0.9144	639	0.7474	688	0.2156	737	0.0452		
591	0.9248	640	0.7257	689	0.2062	738	0.0433		
592	0.9325	641	0.7198	690	0.2009	739	0.0421		
593	0.9389	642	0.7048	691	0.1974	740	0.0405		
594	0.9478	643	0.6911	692	0.1909	741	0.0408		
595	0.951	644	0.6779	693	0.1857	742	0.0382		
596	0.9585	645	0.664	694	0.1781	743	0.0379		
597	0.9678	646	0.6545	695	0.1721	744	0.0359		
598	0.9717	647	0.639	696	0.1686	745	0.035		

6. Goniophotometer Test results for Model # CP05-180L-[D;M;I;S;P;SI;SM;SP;SMP;SIP]-30

6.1 Test Data

Test Ambient Temperature	25.2℃	Test orientation	Downward
Operate time(Min.)	90	stabilization time(Min.)	30

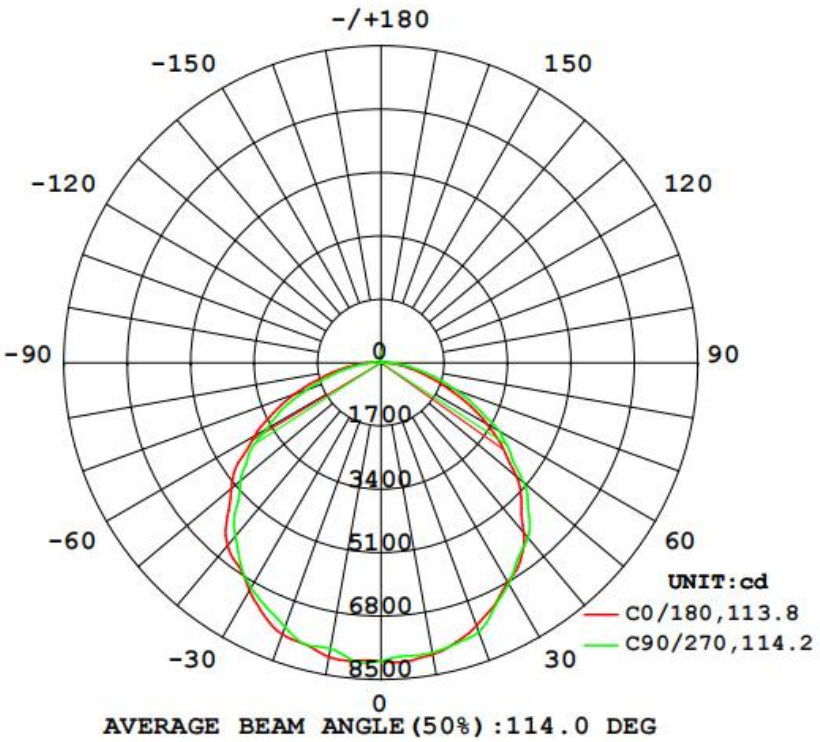
Electrical Measurement

Input Voltage (V)	Frequency (Hz)	Input Current(A)	Power Factor	Power(W)
120.05	60	1.5021	0.9979	179.95

Optical Measurement

Luminous Flux (lm)	Efficacy(lm/W)	Imax (cd)	ZL (0-40°)	ZL (40-70°)
23598.1	131.14	8086	44.1%	46.4%

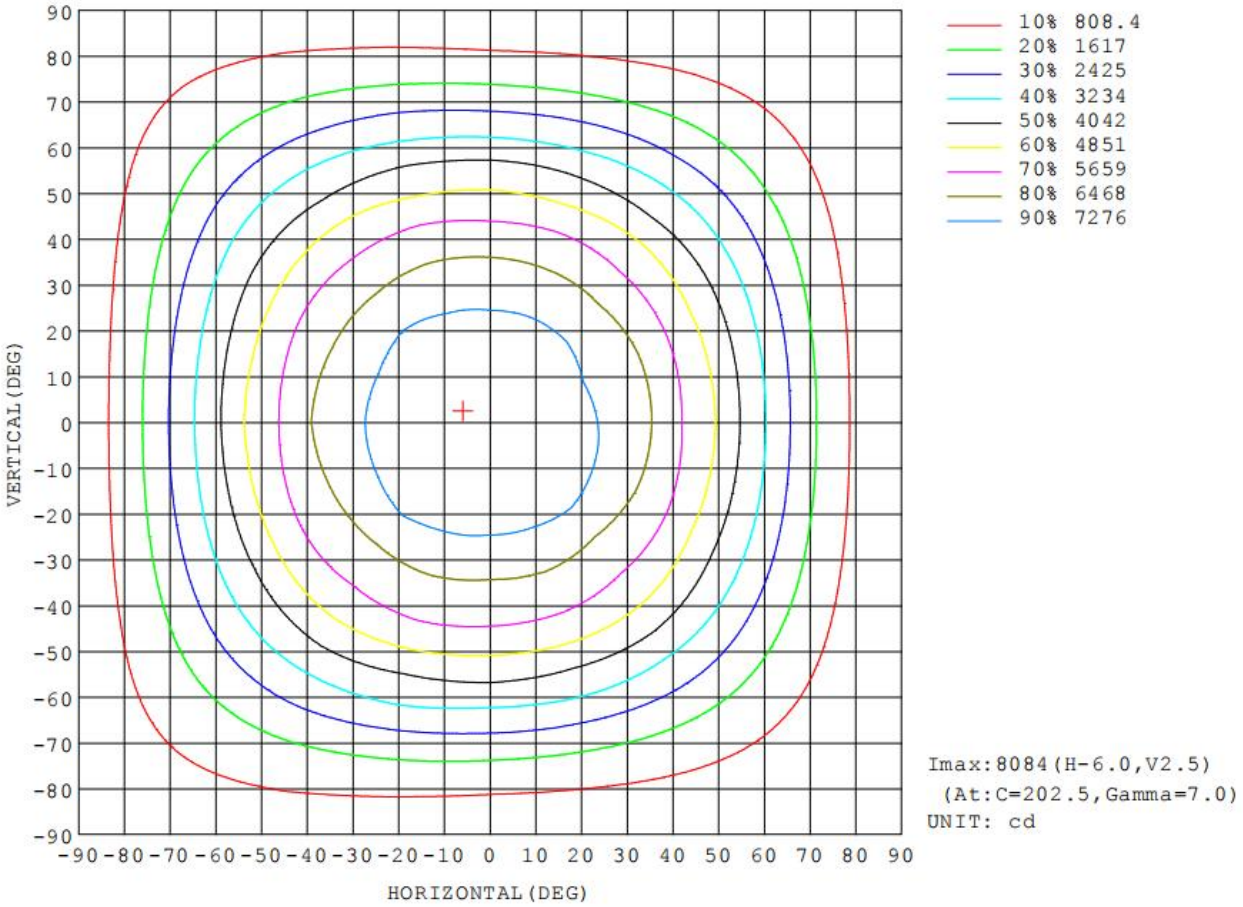
6.2 Luminous Intensity Distribution



6.3 Zonal Flux Diagram

γ	C0	C45	C90	C135	C180	C225	C270	C315	γ	Φ zone	Φ total	%lum,lamp
10	7947	7933	7900	7927	8018	7904	7788	7834	0- 10	760.1	760.1	3.22,3.22
20	7503	7576	7680	7605	7752	7637	7536	7500	10- 20	2197	2957	12.5,12.5
30	6801	6812	6852	7097	7063	7090	6971	6793	20- 30	3361	6319	26.8,26.8
40	5960	5915	6134	6078	6402	6240	6078	5929	30- 40	4088	10407	44.1,44.1
50	4751	4830	5004	5182	5236	5194	4928	4766	40- 50	4296	14703	62.3,62.3
60	3277	3349	3678	3816	3911	3789	3517	3356	50- 60	3821	18524	78.5,78.5
70	1798	1919	2086	2373	2474	2378	2150	1926	60- 70	2828	21352	90.5,90.5
80	686.1	807.5	931.5	1111	1156	1124	931.1	815.3	70- 80	1588	22940	97.2,97.2
90	30.81	65.54	47.70	286.5	300.3	293.1	137.5	27.50	80- 90	572.1	23512	99.6,99.6
100	81.70	18.46	2.046	16.70	32.42	16.08	3.269	23.05	90-100	38.93	23551	99.8,99.8
110	3.340	3.168	2.802	3.374	34.37	4.600	3.681	3.898	100-110	22.31	23573	99.9,99.9
120	4.240	4.101	4.072	3.861	4.187	4.231	4.021	4.094	110-120	4.164	23577	99.9,99.9
130	5.235	5.309	5.452	5.031	4.848	4.752	5.068	5.364	120-130	4.071	23581	99.9,99.9
140	6.413	6.134	6.416	6.070	6.459	6.542	6.919	7.425	130-140	4.511	23586	99.9,99.9
150	6.372	6.645	6.783	6.635	8.694	8.800	9.017	9.386	140-150	4.504	23591	100,100
160	6.843	7.599	7.841	7.548	9.984	10.27	10.72	10.49	150-160	3.901	23594	100,100
170	8.205	8.530	8.761	8.295	10.11	10.12	10.40	10.64	160-170	2.595	23597	100,100
180	316.9	316.9	9.934	9.934	9.934	9.934	316.9	316.9	170-180	1.122	23598	100,100
DEG	LUMINOUS INTENSITY:cd									UNIT:lm		

6.4 Isocandela Diagram



6.5 Luminous Distribution Intensity Data

Table--1 UNIT: cd

γ (DEG) \ C(DEG)	0	22.5	45	67.5	90	112.5	135	157.5	180	202.5	225	247.5	270	292.5	315	337.5			
0	8010	8010	8010	8010	8010	8010	8010	8010	8010	8010	8010	8010	8010	8010	8010	8010			
5	8040	8023	7994	7946	7898	7875	7886	7924	8024	8068	8078	8057	8039	8034	8021	8024			
10	7947	7914	7933	7940	7900	7887	7927	7957	8018	8023	7904	7775	7788	7823	7834	7871			
15	7767	7735	7727	7765	7797	7806	7829	7829	7822	7819	7788	7815	7820	7753	7688	7685			
20	7503	7577	7576	7612	7680	7684	7605	7599	7752	7666	7637	7611	7536	7531	7500	7417			
25	7188	7231	7320	7243	7233	7341	7430	7410	7457	7398	7422	7307	7249	7244	7222	7098			
30	6801	6820	6812	6802	6852	6895	7097	7049	7063	7016	7090	6989	6971	6891	6793	6748			
35	6503	6457	6340	6425	6411	6508	6546	6637	6661	6683	6622	6648	6577	6501	6436	6487			
40	5960	6055	5915	6037	6134	6158	6078	6267	6402	6337	6240	6200	6078	6045	5929	5996			
45	5340	5347	5425	5526	5595	5725	5678	5827	5826	5839	5707	5694	5490	5456	5382	5301			
50	4751	4686	4830	4938	5004	5099	5182	5215	5236	5192	5194	4989	4928	4807	4766	4677			
55	4000	3965	4112	4139	4244	4285	4468	4514	4677	4576	4457	4405	4355	4143	3991	4007			
60	3277	3298	3349	3496	3678	3687	3816	3779	3911	3875	3789	3723	3517	3433	3356	3304			
65	2524	2572	2634	2716	2804	2958	3039	3115	3185	3197	3112	3010	2884	2749	2628	2587			
70	1798	1861	1919	2009	2086	2255	2373	2425	2474	2457	2378	2272	2150	2006	1926	1834			
75	1185	1229	1299	1388	1488	1583	1674	1734	1746	1742	1719	1602	1455	1376	1299	1235			
80	686	739	808	867	931	1043	1111	1144	1156	1159	1124	1053	931	870	815	741			
85	319	376	422	439	475	576	659	675	667	691	672	588	471	446	416	368			
90	30.8	44.8	65.5	71.4	47.7	195	286	317	300	323	293	191	138	77.6	27.5	14.4			
95	11.4	22.9	6.25	2.00	1.52	1.80	2.20	24.2	4.63	7.53	3.08	2.61	2.66	3.25	9.35	17.1			
100	81.7	60.3	18.5	4.12	2.05	4.40	16.7	61.9	32.4	56.2	16.1	5.61	3.27	5.67	23.0	66.6			
105	33.7	16.6	3.36	2.53	2.41	3.02	13.7	59.9	86.4	61.5	14.9	4.32	3.64	3.68	4.47	19.9			
110	3.34	3.48	3.17	2.89	2.80	2.80	3.37	16.0	34.4	16.3	4.60	3.86	3.68	3.70	3.90	4.16			
115	3.71	3.79	3.61	3.39	3.35	3.27	3.32	3.63	4.05	4.37	4.23	3.88	3.75	3.75	3.91	4.11			
120	4.24	4.29	4.10	4.03	4.07	3.87	3.86	4.02	4.19	4.35	4.23	3.97	4.02	3.98	4.09	4.26			
125	4.76	4.78	4.61	4.65	4.77	4.49	4.44	4.49	4.26	4.46	4.27	4.17	4.41	4.33	4.42	4.70			
130	5.24	5.20	5.31	5.42	5.45	5.23	5.03	5.17	4.85	4.90	4.75	4.87	5.07	5.18	5.36	5.34			
135	5.81	5.80	5.67	5.83	6.01	5.86	5.65	5.59	5.49	5.47	5.63	5.75	5.92	6.08	6.30	6.47			
140	6.41	6.26	6.13	6.20	6.42	6.26	6.07	6.13	6.46	6.51	6.54	6.69	6.92	7.11	7.42	7.54			
145	6.24	6.21	6.26	6.34	6.60	6.57	6.44	6.62	7.63	7.61	7.70	7.82	7.98	8.19	8.53	8.45			
150	6.37	6.53	6.64	6.64	6.78	6.80	6.63	6.81	8.69	8.64	8.80	8.89	9.02	9.12	9.39	9.26			
155	6.67	7.11	7.31	7.31	7.35	7.29	7.07	7.22	9.37	9.57	9.73	9.94	10.00	9.98	10.2	9.93			
160	6.84	7.38	7.60	7.68	7.84	7.75	7.55	7.82	9.98	10.1	10.3	10.5	10.7	10.6	10.5	10.2			
165	7.61	7.91	8.07	8.14	8.17	8.02	7.79	8.07	10.1	10.2	10.3	10.4	10.7	10.6	10.6	10.3			
170	8.20	8.36	8.53	8.70	8.76	8.60	8.29	8.55	10.1	10.1	10.1	10.2	10.4	10.5	10.6	10.4			
175	9.34	9.48	10.0	10.3	9.42	9.10	8.83	9.15	9.85	9.83	9.81	9.87	9.96	10.1	10.3	10.3			
180	317	317	317	317	9.93	9.93	9.93	9.93	9.93	9.93	9.93	9.93	317	317	317	317			

7. THD and PF Test

Model	Voltage (V AC)	Frequency (Hz)	Power Factor	THD (%)
CP05-180L-[D;M;I;S;P;SI; SM;SP;SMP;SIP]-30	120.0	60	0.998	2.45
	277.0	60	0.951	9.05
CP05-180L-[D;M;I;S;P;SI; SM;SP;SMP;SIP]-65	277.0	60	0.951	9.1

8. Photo of sample

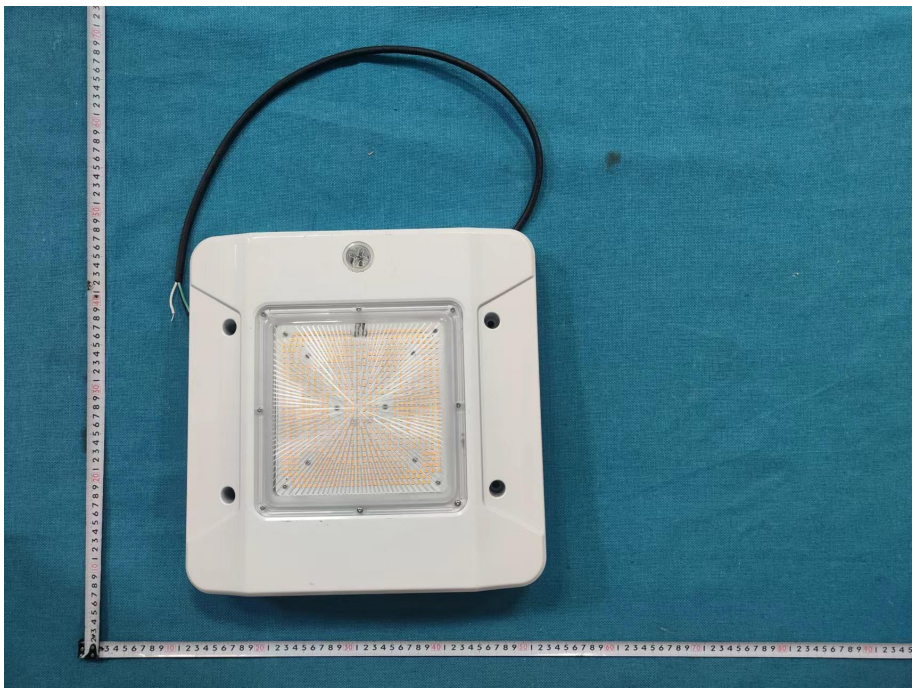


Figure 1

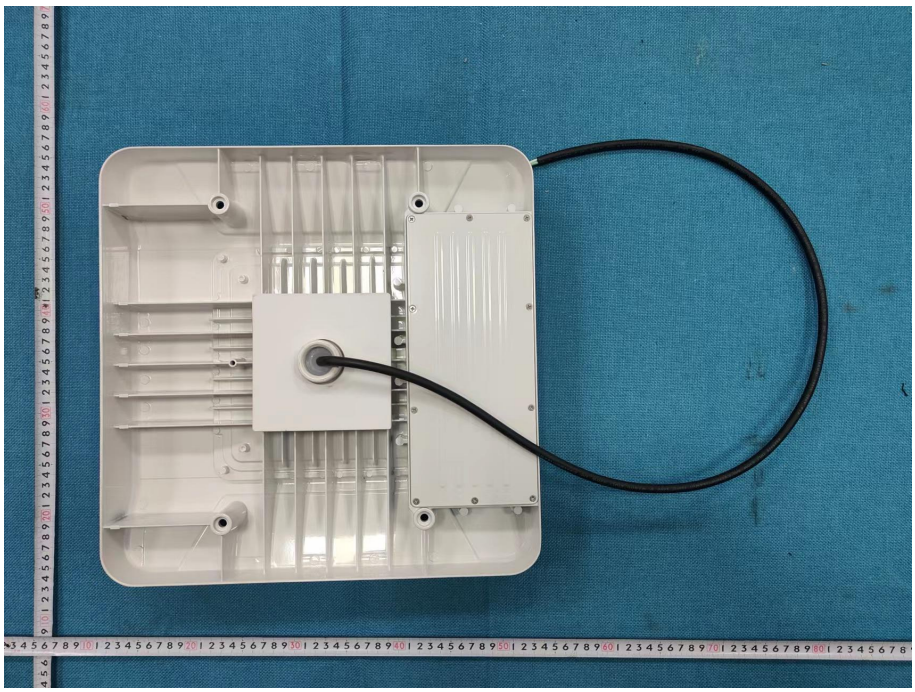


Figure 2

---End of Report---